CT · WEEK 2

CT week 2 patterns: Filtering, AND / OR, max and min

Pseudocode, filtering & AND / OR · Max, min & 'who has it'. Save this PDF for revision.

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Exam-style pseudocode

```
count = 0
while(Table 1 has more rows){
    Read the first row X in Table 1
    if(X.Gender == "F" and X.CityTown == "Chennai"){
        count = count + 1
    }
    Move X to Table 2
}
```

count = number of female students from Chennai.

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Operators

Write	Means
a = b	Assign: store the value of b in a
a == b	Compare: is a equal to b?
a != b	Not equal
and	True only when both sides are true
or	True when at least one side is true
not	Flips True and False
a mod b	Remainder: 17 mod 5 = 2

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Logic you'll use

De Morgan

not (A and B) = (not A) or (not B)

De Morgan

not (A or B) = (not A) and (not B)

Count of A or B

count(A) + count(B) − count(A and B)

Exactly one of A, B

(A and not B) or (B and not A)

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Key points

- Brackets change meaning: (A and B) or C ≠ A and (B or C).
- Check whether each update sits inside or outside the if. Indentation and braces decide.
- A = A + 1 counts cards. A = A + X.Marks adds up a field.

MAX, MIN & 'WHO HAS IT' · IN EVERY PAPER

Highest mark and who scored it

```
max = 0
topper = "None"
while(Table 1 has more rows){
    Read the first row X in Table 1
    if(X.Mathematics > max){
        max = X.Mathematics
        topper = X.Name
    }
    Move X to Table 2
}
```

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Filtered min (PYQ May 2023: lowest Chemistry, male, Vellore)

```
min = 100
while(Table 1 has more rows){
    Read the first row X in Table 1
    if(X.Gender == "M" and X.CityTown == "Vellore"){
        if(X.Chemistry < min){
            min = X.Chemistry
        }
    }
    Move X to Table 2
}
```

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Ties: which card wins?

Comparison	On a tie, keeps
X.M > max	the first card with the maximum
X.M >= max	the last card with the maximum
X.M < min	the first card with the minimum
X.M <= min	the last card with the minimum

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Key points

- max = 0 is safe only when every value is ≥ 0. For min, start above the largest possible value (100 for marks) or with the first card's value.

- Second highest: if X > max { second = max; max = X } else if X > second { second = X }.

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Common traps

⚠ min = 0 bug: no mark is below 0, so min never changes. A favourite 'find the mistake' option.

⚠ Setting topper outside the if stores the last card's name, not the topper's.

⚠ If no card passes the filter, max and min keep their starting values.

Notes for every week of CT, free.

Full notes, the pattern sheet and common traps, with worked PYQs.

btechi.in/qualifier/notes/ct →

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